312337

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions :

(1) All Questions are *compulsory.*

(2) Answer each next main Question on a new page.

(3) Illustrate your answers with neat sketches wherever necessary.

(4) Figures to the right indicate full marks.

(5) Assume suitable data, if necessary.

(6) Use of Non-programmable Electronic Pocket Calculator is permissible.

(7) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

(8) Use of steam tables, logarithmic, Mollier's chart is permitted.

Marks

1. Attempt any FIVE : **10**

(a) Enlist any four objectives of sizing.

(b) Write the importance of scouring of cotton for subsequent wet processing.

(c) Enlist any four bleaching agents used in bleaching of textiles.

(d) State the distinct stages involved in the mercerisation of cotton.

(e) Enlist the factors affecting the mercerisation process.

(f) Draw a labelled sketch of a yarn mercerisation machine.

(g) Enlist any four changes occurring in cotton due to mercerisation.

2. Attempt any THREE : **12**

(a) A fabric roll of 240 yards long and 48 inch width contains following defects using 4 point system :

(i) Defect upto 3″ = 04

(ii) Defect over 3″ but less than 6″ = 10

(iii) Defect over 6″ but less than 9″ = 02

(iv) Defect over 9″ = 02

If the customer's criteria for acceptability is 40 points / 100 sq. yards, comment on the acceptability of the inspected lot.

(b) Elaborate the role of different ingredients used in the size paste formulation with one suitable exmaple.

(c) Elaborate the mechanism of scouring process with suitable chemical reactions.

(d) Explain with reactions, the mechanism of bleaching cotton fabric using sodium hypochlorite.

3. Attempt any THREE : **12**

(a) Elaborate the classification of softeners used in sizing of textile warp yarns.

(b) Describe the Kier boiling method of scouring 100% cotton voile fabric with the half of a labelled sketch.

(c) Compare Hydrogen peroxide bleaching with sodium hypochlorite bleaching agents (any four points).

(d) Propose the significance of causticization and elaborate the process.

4. Attempt any THREE : **12**

(a) Compare pad-chainless and padless– chainless mercerisation (any four points).

(b) Elaborate the bleaching procedure for wool by using hydrogen peroxide.

(c) Explain the absorbency method to evaluate the scouring efficiency of 100% cotton fabric.

(d) Differentiate between enzymatic desizing and conventional desizing (any four points).

(e) Describe the construction and working of a shearing and cropping machine with a neat labelled sketch.

5. Attempt any TWO : 12

(a) Propose a machine with a neat labelled sketch, used for singing of 100% cotton poplin fabric. Also outline the procedure of the same.

(b) Evaluate the desizing efficiency of 100% cotton by weight loss method. Also propose the factors affecting the desizing process.

(c) Describe the scouring method adopted to remove the hydrophobic impurities from

(i) Grey wool fabric

(ii) Raw silk fabric

6. Attempt any TWO : 12

(a) Propose a size paste formulation for

(i) 100% cotton shirting

(ii) 100% polyester fabric

(iii) Polyester / Cotton suiting blend

(b) Outline and elaborate the factors affecting bleaching of 100% cotton using hydrogen peroxide.

(c) Propose the following methods to evaluate the efficiency of mercerisation process :

(i) BAN

(ii) Axial ratio